

Causal Structure Discovery via a Self-Implemented FCI⁺ Algorithm

A Case Study on the Tennessee STAR Project

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Abstract. This project independently implements paper-aligned FCI and FCI⁺, validates them against exact graphical oracles and established software, and separates algorithm fidelity from a robust finite-sample application workflow. Latent common causes and selection can make local adjacency conditioning insufficient, while standard FCI's Possible-D-SEP search is expensive and sample conditional-independence (CI) errors propagate into PAG endpoints. The implementation therefore provides both the historical search schedule of Spirtes et al. (2000) and Algorithm 2 of Claassen et al. (2013), whose $O(N^{2(k+2)})$ result is a CI-query bound under a fixed true observed-MAG degree bound k , faithfulness, and an exact constant-time oracle. The implementation is delivered as a self-contained, fully typed Python package of roughly 9,600 lines across 38 modules, written without depending on any existing causal-discovery library and validated by 367 automated tests on a five-version CI matrix (Python 3.9–3.13). Validation proceeds in layers: unit and property tests; exact MAG m-separation fixtures; the published Figure 4(b) MAG and its exhibited separator $\{U, V, Z\}$; a 15-run known-truth benchmark; executed differential comparisons with causal-learn 0.1.4.7 and pcalg 2.7-12; and finite-sample order, bootstrap, sensitivity, and chronology audits. On the committed known-truth benchmark, robust FCI⁺ attains mean skeleton F1 0.985, semantic-edge F1 0.985, and endpoint accuracy 0.840, against 0.977, 0.943, and 0.737 for paper-profile FCI, at a mean logical CI count of 402 rather than 450. These results describe the tested cases and policies; they do not establish general statistical dominance. The separate Tennessee STAR application analyses 6,325 kindergarten-cohort students from the official 11,601-record dataset. Its strongest evidence is a design-supported observed-arm contrast: small-class assignment shows 13.90 points higher observed kindergarten achievement than regular-class assignment (school-cluster 95% interval 5.21 to 22.07), whereas the regular-with-aide contrast is near zero. Because that outcome subset is post-assignment selected and the original randomization blocks and compliance histories were not reconstructed, the contrast is consistent with a beneficial early effect rather than an identified selected-subset causal effect. Across discovery methods, skeleton agreement is more defensible than several order-, threshold-, or chronology-sensitive endpoint marks; no PAG in this report identifies a unique DAG or estimates a treatment effect.

Keywords: causal discovery; FCI; FCI⁺; partial ancestral graph; latent confounding; conditional independence; Tennessee STAR.

Scope of this document. This is the standalone short report. A companion full report (`reports/fci_plus_star_report.tex`) retains the complete page-level source locators, the per-stage concept/symbol/test matrix, the reproducibility appendix, and the extended result tables. Every number below is regenerated from committed artifacts by the scripts listed in section 10.

1 Introduction

Many empirical studies ask causal questions in settings where not every common cause has been measured. Family background, institutional selection, health, motivation, and administrative recording processes can all generate dependence that cannot be represented honestly by a causally sufficient DAG over the recorded columns. Fitting a directed graph while silently assuming causal sufficiency turns uncertainty into unjustified arrows.

FCI was designed for this problem. Under causal Markov, faithfulness, and appropriate sampling assumptions, it uses conditional-independence information to learn a Partial Ancestral Graph (PAG) that remains valid when latent confounders and selection variables may exist (Spirtes et al., 2000; Richardson and Spirtes, 2002). The price of the broader model class is additional search and greater endpoint ambiguity: standard FCI’s Possible-D-SEP refinement may test many separating sets that lie outside local adjacency neighbourhoods. Claassen et al. (2013) address the search problem for sparse graphs with a logical characterisation of D-SEP links and a hierarchical procedure for finding the relevant separating sets, yielding a polynomial worst-case query bound for a fixed maximum degree k .

Key result. This project independently implements paper-aligned FCI and FCI⁺, validates them against exact graphical oracles and established software, and separates algorithm fidelity from a robust finite-sample application workflow.

That proposition is delivered as four concrete, auditable contributions.

- C1. *Algorithm implementation.*** A self-contained, fully typed Python package (`fci_engine`; 9,559 lines across 38 modules, written from the primary sources without depending on any existing causal-discovery library) implementing both search schedules end to end: five conditional-independence tests, an explicit PAG endpoint algebra, the ordered Possible-D-SEP stage, the complete Algorithm 2 augmented-skeleton and hierarchy stage, and a full Zhang-style orientation engine with five selectable rule schedules.
- C2. *Validation apparatus.*** An exact MAG m -separation oracle, encoded published reference fixtures, a 15-run known-truth benchmark, and 367 automated tests in 44 modules (5,929 lines), executed under strict MyPy typing and a five-version CI matrix (Python 3.9–3.13).
- C3. *Software positioning.*** Executed differential comparisons against causal-learn and R `pcaI`g under matched CI semantics. Among the public interfaces checked on 26 July 2026, no other Python implementation of Claassen’s FCI⁺ with paper-profile control and a structured audit trail was identified, so the package fills a documented gap rather than duplicating an existing tool.
- C4. *Application.*** A separate 3,052-line Tennessee STAR case study (plus a 396-line independent R replication runner) that consumes only the public API, with school-cluster bootstrap, cyclic-order, chronology, and sensitivity audits organised by a pre-specified evidence hierarchy.

The separation between C1 and C4 is central to the research design: an algorithm should remain reusable and domain-neutral, whereas an application must state its own coding, selection, measurement, clustering, and interpretation decisions.

The report addresses five questions. **RQ1:** how do the original FCI and FCI⁺ sources define their targets, assumptions, search stages, and guarantee boundaries? **RQ2:** does the implementation map those stages to explicit code, public profiles, and tests? **RQ3:** what is recovered under exact-oracle and finite-sample known-truth conditions? **RQ4:** how does the package compare fairly with established FCI software in executed benchmarks and documented capabilities? **RQ5:** which STAR conclusions are design based, robust at the skeleton level, unresolved at the endpoint level, or merely exploratory? Sections 3 and 4 address theory and implementation, sections 5 and 6 assess controlled evidence, and sections 7 to 9 form a separate application. This ordering prevents a real-data pattern from being used as evidence that the algorithm matches its sources.

2 Graphical Background

Let V be the observed variables, L latent variables, and S selection variables. Constraint-based discovery begins from statements $X \perp Y \mid Z$ with $Z \subseteq V \setminus \{X, Y\}$. Two assumptions link a causal graph to a distribution. The *causal Markov condition* states that graphical separation implies probabilistic conditional

independence. *Faithfulness* states that observed independences arise from graphical separation rather than exact cancellation of causal effects.

Under causal sufficiency these assumptions support DAG-based methods such as PC. When latent or selection variables are present, the appropriate observed-variable object is a maximal ancestral graph (MAG) (Richardson and Spirtes, 2002): marginalising latent variables and conditioning on selection variables induces directed, bidirected, and undirected relations among observed nodes, and every missing MAG edge corresponds to an m-separation. Different MAGs may encode the same independence model, so the identifiable output is a PAG, which records exactly the endpoint features shared by the MAGs of one observational Markov equivalence class (Zhang, 2008).

M-separation generalises d-separation to mixed ancestral graphs and is the criterion used by the exact graphical oracle in this project. A sample Fisher-Z or G-square test, by contrast, is an error-prone statistical decision. Oracle recovery therefore isolates graph logic and cannot be transferred directly to finite-sample accuracy.

Table 1: Interpretation of common PAG edge marks.

Mark	Interpretation
$X \text{ o-o } Y$	The adjacency is retained, but neither ancestral endpoint is identified.
$X \text{ o-} \rightarrow Y$	The arrowhead at Y excludes Y as an ancestor of X ; the endpoint at X remains unresolved.
$X \rightarrow Y$	The tail-arrowhead pattern supports X as an ancestor of Y through some path in every represented MAG. It does not assert a direct structural coefficient.
$X \leftrightarrow Y$	Both endpoints are arrowheads. The pattern is compatible with latent confounding, but it does not identify a unique latent variable and can be distorted by selection or CI-test error.
$X - Y$	An undirected ancestral-graph edge, often associated with selection structure.

An adjacency says only that the measured variables were not separated by the tested sets under the recorded policy. A tail or arrowhead constrains ancestry across the equivalence class. Neither fact alone identifies a direct structural coefficient, an adjustment set, or an interventional contrast. These distinctions determine the evidence hierarchy used in section 9.

3 The Two Source Algorithms

3.1 Standard FCI and its historical boundary

FCI reasons from conditional independences in the marginal distribution over measured variables O ; it does not require that all common causes of measured variables are themselves measured (Spirtes et al., 2000, Ch. 6). A PC-style adjacency search removes $X - Y$ when some subset of the current local neighbours separates the pair. With latent variables, a pair may remain adjacent after local search even though it is separable over the observed variables, because a necessary separator can contain a node that is no longer adjacent to either endpoint by the time the search reaches the relevant depth.

The book introduces D-SEP to characterise where an observed separator can reside, and, because D-SEP is not directly available from a partially learned graph, constructs the observable superset Possible-D-SEP. For an ordered pair (A, B) , a node $V \neq A$ belongs to Possible-D-SEP(A, B) when there is a path from A to V whose every internal triple has a collider middle node, or has a middle node that is not a definite noncollider while the two outer nodes are adjacent. The triangle clause is not a claim that the middle node is a true collider; it retains a node when the current marks do not yet justify excluding it. The construction is deliberately permissive, and it is required *separately* from each ordered endpoint: a valid separator may be available from one direction but not the other, and an implementation that uses one direction only, or that mixes members of the two pools, is no longer executing the same search contract.

The full schedule is: complete graph and PC-style adjacency search; an initial unshielded-collider phase;

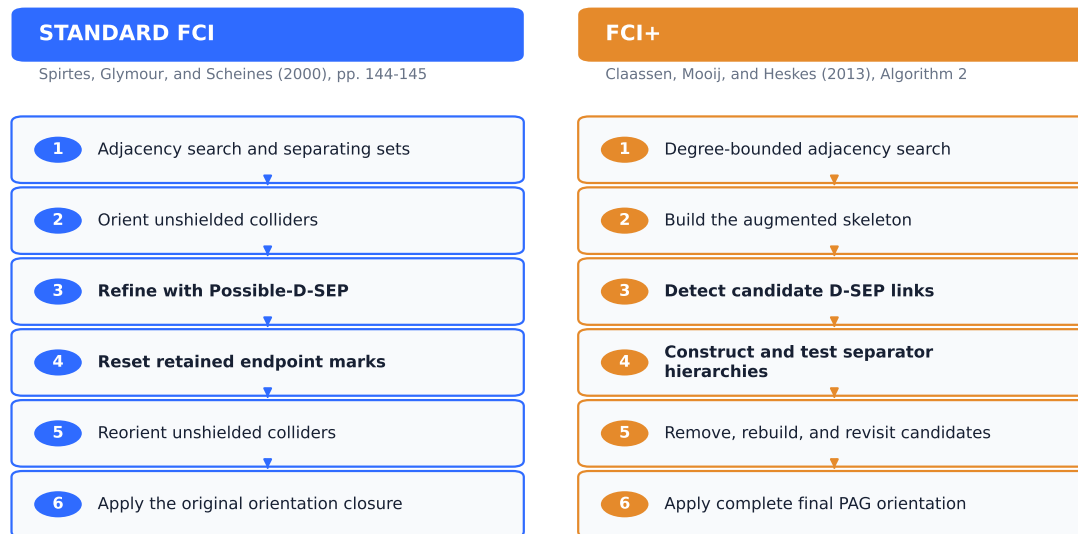
ordered Possible-D-SEP construction and non-local CI refinement; endpoint reset and collider reconstruction from the final separator information; and orientation closure. Theorem 6.4 states that, under faithfulness and correct population-level independence decisions, this returns a correct *partially oriented inducing path graph* for the observed margin. The sentence immediately following that theorem is equally important: the authors state that they did not know whether the output was always maximally informative. Complete orientation of all invariant marks therefore depends on the later rule set of Zhang (2008), and this report does not retroactively attribute modern PAG completeness to the 2000 source.

3.2 FCI+: Algorithm 2

Claassen et al. (2013) ask whether causal discovery remains intrinsically exponential when the observed structure is sparse but latent variables and selection bias may be present. Their C-LEARN task assumes an independence oracle answering any query exactly in constant time, a distribution faithful to an underlying causal DAG, an observed-variable MAG whose maximum node degree is bounded by a constant k , and a completed PAG as the output target. Under those assumptions they establish the worst-case CI-query bound $O(N^{2(k+2)})$, where N is the number of observed variables. The claim is narrower than the paper’s memorable title: it is a polynomial query-complexity result for a sparse, faithful oracle problem, and it does not make a statistical CI test exact or constant-time.

After a bounded adjacency search, a nonadjacent pair in the true MAG can remain falsely connected because every useful separator requires a node outside the current endpoint-adjacency union; such a surviving false adjacency is a *D-sep link*. Rather than searching every subset of a large Possible-D-SEP pool, FCI+ extracts additional invariant non-ancestry information into an *augmented skeleton* G^+ : if a known minimal independence $X \perp Y \mid S$ becomes a dependence $X \not\perp Y \mid S \cup \{A\}$ for a suitable adjacent node A , an invariant arrowhead at A is supported on the relevant incident edges. In G^+ , a D-sep link $X - Y$ appears as the middle edge of $U \leftrightarrow X \leftrightarrow Y \leftrightarrow V$ with U and V nonadjacent and the required cross-direction paths not running against an arrowhead. This pattern only *proposes* a candidate; independence evidence still decides deletion.

The conditioning set comes from the recursive hierarchy $\text{HIE}(X, I)$, the fixed point of $H_{t+1} = H_t \cup \bigcup_{A, B \in H_t, A \neq B} \text{Sepset}_I(A, B)$, where I is the collection of recorded minimal independence statements. Candidate links can depend on one another, so a successful removal changes the independence information and requires the augmented skeleton and the candidate list to be rebuilt. The query accounting sums a bounded adjacency search ($O(N^{k+2})$), augmentation ($O(N^3)$), at most $O(N^2)$ candidates, $O(N^{2k})$ paired endpoint bases, $O(N^2)$ minimisation queries, and an $O(N^2)$ revisit factor, giving the dominant $O(N^2) O(N^{2k}) O(N^2) = O(N^{2(k+2)})$. The arXiv proof supplement’s abstract states $N^{2(k+1)}$, while its theorem, proof, and concluding sentence agree with the UAI paper on $N^{2(k+2)}$; the discrepancy is recorded rather than silently harmonised. Finally, Algorithm 2 invokes complete FCI orientation rather than rederiving it, so the completed-output claim inherits Zhang (2008) as a dependency.



Shared target: an observational equivalence-class representation; neither workflow identifies a unique DAG or a treatment effect.

Figure 1: Source-located control-flow comparison. Standard FCI performs a non-local Possible-D-SEP refinement; FCI+ instead iterates augmented-skeleton, candidate D-sep-link, hierarchy, and minimal-separator stages before invoking final orientation. The robust application profile of section 4 changes finite-sample policy, not the definition of Algorithm 2.

3.3 What the sources do not establish

Neither source guarantees finite-sample recovery: both theorems are conditional on correct CI decisions, and no bound is given for Fisher-Z, G-square, kernel, or missing-data tests at a given N . Neither identifies a unique DAG, an adjustment set, or a treatment effect; a bidirected endpoint pattern does not name, count, or measure a hidden cause. The 2000 source proves no maximal orientation and gives no fixed-degree polynomial bound. The 2013 source proves no finite-sample superiority over standard FCI, no wall-clock dominance, and no guarantee when k is wrong or growing; a user-entered sparsity bound is not evidence that the assumption holds. Claassen et al. (2013) also warn that hierarchy-derived conditioning sets may be large and may reduce finite-sample power.

4 Independent Python Implementation

4.1 Package boundary and public API

The objective is not to relabel a generic constraint-based search as FCI+, but to expose the algorithmic obligations of section 3 as inspectable components and keep them separate from finite-sample application policy. Reusable code resides in `src/fci_engine`; the STAR workflow resides in `case_studies/tennessee_star`. No STAR variable, temporal constraint, discretisation choice, or substantive conclusion is embedded in the estimator package.

The top-level functions keep the two algorithms as separate entry points, and paper profiles are explicit because a default optimised for finite samples should not silently stand in for the source algorithm. `FCIConfig.paper()` fixes unbounded depth and path length, immediate skeleton deletion, first-found separators, enabled Possible-D-SEP, nonconservative colliders, and the historical `spirtes_2000` orientation closure. `FCIPlusConfig.paper(k)` requires an explicit k , assigns it to both the initial conditioning depth and the endpoint-base sparsity bound, disables ordinary Possible-D-SEP, uses immediate deletion and first-found separators, and selects the complete standard orientation schedule required by Algorithm 2's output claim. Profiles constrain execution; they do not validate theorem assumptions. Supplying $k = 3$ does not prove that the unknown true MAG has degree at most three.

The PAG class stores an endpoint mark at each side of every adjacency while exposing the adjacency once, so tails, arrowheads, circles, and absence are represented explicitly rather than flattened into a directed graph. Mutations validate both nodes and reject one-sided or NONE-marked edges. Every search uses one `CITest.test(data, x, y, cond_set)` contract, with built-in Fisher-Z, missing-value Fisher-Z, chi-square, G-square, and kernel CI tests. A cache canonicalises the unordered pair and conditioning set, so the result separates the number of *logical* CI requests made by control flow, the subset answered from cache, and the underlying statistical outcomes. This distinction matters when connecting measurements to the FCI⁺ theorem, which counts oracle queries rather than elapsed time.

4.2 Pipelines, orientation strategies, and profiles

FCI. `fit` exposes the source stage ordering directly: complete circle graph and adjacency search; separator-based collider decisions; ordered Possible-D-SEP refinement recording removals with source `pdsep`; endpoint reset and collider reconstruction; then the configured closure. FCIPlus. `fit` deliberately does not call Possible-D-SEP; it passes the initial skeleton and separators to the Algorithm 2 search, which builds the augmented skeleton, recognises literal bidirected witness patterns, enumerates paired endpoint bases, expands each hierarchy seed to a fixed point, tests the resulting conditioning set, minimises the separator, updates the graph with source `fci_plus_dsep`, and rebuilds candidates. A diagnostics record counts candidate sightings and revisits, hierarchy queries and cache hits, duplicate conditioning-set skips, D-SEP-stage CI requests, removed edges, and the maximum tested conditioning size. These counters make a run inspectable; they are empirical accounting, not a mechanised proof of the asymptotic bound.

One dispatcher provides explicit orientation schedules, and the separation is part of the scientific interface. `spirtes_2000` is the original arrowhead and discriminating-path closure used by the standard-FCI paper profile, historically aligned but not claimed maximally informative. `standard` is the modern complete schedule including later tail-completion rules, the appropriate completeness dependency for the FCI⁺ paper profile. `conservative` retains arrowhead conclusions while suppressing tail-producing propagation; `leaf` permits a restricted tail rule when the newly directed endpoint is a leaf; and `robust` combines conservative collider decisions with the leaf-tail schedule. The last three are finite-sample engineering policies and support no completed-PAG claim. A dedicated regression verifies that the `spirtes_2000` profile stops before a selection-bias tail completion that `standard` can orient, making the historical completeness boundary executable rather than merely descriptive.

Table 2: Published concept, repository symbol, and executable validation. A “paper mechanism” classification means the symbol implements the stated source operation when invoked through the corresponding paper profile; it does not mean the source proves every line of Python correct.

Published concept	Repository symbol	Principal validation	Classification
PC-style adjacency search	<code>skeleton.learn_initial_skeleton</code>	skeleton, profile, and reference integration tests	Shared
Possible-D-SEP criterion	<code>pdsep.possible_dsep</code>	collider/triangle, tail blocking, cycles, path bounds	Spirtes
Ordered Possible-D-SEP search	<code>pdsep.refine_skeleton_with_pdsep</code>	both-direction and no-mixed-pool tests	Spirtes
Original orientation closure	<code>rules.apply_orientation_rules(spirtes_2000)</code>	orientation-rule tests; historical stopping regression	2000-aligned, incomplete by source
Complete modern orientation	<code>rules.apply_orientation_rules(standard)</code>	<code>test_oracle_pag_rules.py</code>	Later Zhang dependency
Augmented skeleton	<code>dsep.build_augmented_skeleton</code>	single-node-dependence and existing-edge tests	Claassen
Candidate D-sep pattern	<code>dsep.possible_dsep_links</code>	witness, cross-path, arrowhead, circle-rejection tests	Claassen
Recursive hierarchy	<code>dsep.hierarchy</code>	fixed-point and excluded-pair cache-key tests	Claassen
Separator minimisation and revisit	<code>dsep.minimal_dsep;</code> refinement loop	minimisation, removal, failed-candidate revisit tests	Claassen
Exact m-separation oracle	<code>simulation.MAGOracleCITest</code>	<code>test_oracle_graphs.py</code>	Validation apparatus
Stable deletion, max- p separators	<code>FCIPlusConfig.practical</code>	configuration, order, finite-sample benchmark tests	Engineering extension
Structured trace and exports	<code>result.FCIResult</code>	<code>test_result_exports.py</code>	Engineering extension

`FCIResult` binds the final graph to the evidence that produced it: the resolved configuration, sample and node counts, separators and their stage sources, logical CI count, cache hits, named CI and orientation traces, ambiguous triples, D-SEP diagnostics, and elapsed time. `explain_edge(x, y)` returns the final marks, the separator and its provenance when removed, all direct CI events for the pair, and the attached orientation events; `save_artifacts` exports JSON, a semantic endpoint CSV, and standalone HTML. This prevents a plotted arrow from becoming detached from the profile, CI test, threshold, or orientation decision that created it. The package is typed and ships `py.typed`; the quality workflow runs Pytest, Ruff, strict MyPy, wheel construction, installation, and an import smoke test. Those checks establish interface and packaging consistency; they do not validate faithfulness, true sparsity, CI-test model assumptions, or an empirical causal conclusion.

The practical profile does not claim that uncertainty has been eliminated. Stable deletion reduces one source of variable-order dependence; max- p separator selection spends more CI queries to retain stronger sample independence evidence at a fixed depth; conservative collider and tail policies preserve circles where sample evidence is ambiguous; and automatic α is a documented sample-size heuristic. Each choice can improve a predefined robustness metric while weakening informativeness or increasing computation. Accordingly, “paper FCI+” is the fidelity and query-efficiency subject, and “robust FCI+” is the preferred engineering policy for the reported application because it meets the stated audit criteria while returning less-specific endpoints. That preference is local to the evidence and criteria in this project; it is not a universal accuracy ranking.

4.3 Engineering scope

Table 3 summarises the committed code base, all of it written for this project. Line counts are raw file lengths including docstrings and blank lines; the algorithm package alone contains 8,323 non-blank, non-comment lines. The package exports 47 public symbols, ships `py.typed`, and passes Ruff, strict MyPy, a wheel build, and an installed-import smoke test, with Pytest executed independently on Python 3.9 through 3.13. Every number in this report can be regenerated from the committed artifacts by the scripts in the last row and section 10.

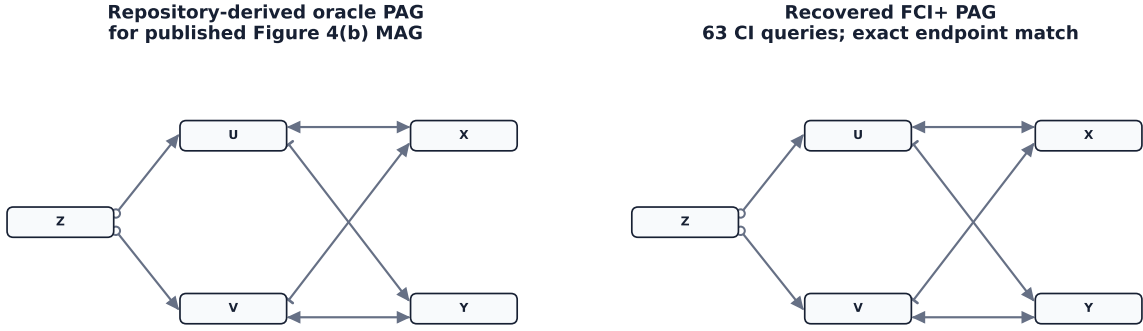
Table 3: Scale of the committed implementation and validation code.

Component	Contents	Lines
Algorithm package <code>src/fci_engine</code>	38 typed modules: PAG endpoint algebra; five CI tests with canonical caching; skeleton, Possible-D-SEP, and Algorithm 2 searches; orientation engine; accuracy and stability metrics; MAG-oracle simulation; result and report exports	9,559
of discovery/ <code>rules.py</code>	which complete arrowhead, discriminating-path, and R5–R10 tail rules under five strategy schedules	1,422
Test suite <code>tests/</code>	367 collected tests in 44 modules: unit, exact-oracle, property, integration, typing, and report regressions	5,929
STAR application	cohort coding, three-panel discovery suite, school-cluster bootstrap, cyclic-order audit, standalone HTML report	3,052
Independent R reference	<code>pcalg: : fciPlus</code> runner with matched G-square semantics	396
Benchmarks, examples, and figure generators	deterministic scripts behind every reported table and figure	6,356
Total		≈25,300

5 Algorithm Validation

5.1 Exact oracle tests

The strongest algorithm tests replace sample-based CI decisions with exact m-separation in a known MAG, isolating graph-search correctness from sampling error. The regression suite contains three independently specified targets based on published structures: the canonical Figure 4(b) D-SEP example of Claassen et al. (2013); a latent-variable reference graph derived from the FCI literature; and an orientation-rule reference graph associated with the completeness results of Zhang (2008). Both FCI and FCI⁺ recover the specified target PAGs on all three encoded fixtures, and Figure 4(b) is additionally tested under four variable orders. The Figure 4(b) MAG has the six edges $Z \rightarrow U$, $Z \rightarrow V$, $U \leftrightarrow X$, $V \rightarrow X$, $U \rightarrow Y$, and $V \leftrightarrow Y$. The adjacency stage retains a false $X - Y$ link, while $X \perp Y \mid \{U, V, Z\}$ and Z is adjacent to neither endpoint. The source exhibits this separator and identifies Z as the D-sep node; it does not print an endpoint-complete PAG, so the completed comparison target is repository derived and is described as such.

Exact m-separation isolates algorithmic correctness from sampling error

The X-Y candidate is removed by the hierarchical D-SEP stage using the separating set $\{U, V, Z\}$.

Figure 2: Exact-oracle validation on the central Figure 4(b) example. The source-grounded facts are the six MAG edges and the exhibited separator; the endpoint-complete comparison target is independently derived in the repository.

Table 4: Paper-aligned Figure 4(b) oracle comparison.

Algorithm	CI queries	Difference	Final PAG
Standard FCI	102	reference	Exact
FCI ⁺	63	-39 (-38.2%)	Exact

These values are frozen in reports/data/oracle_validation_summary.csv together with the target PAG hash, the exact-oracle profile, query counts, separator provenance, endpoint metrics, and execution status.

5.2 Finite-sample behaviour

Real data replace exact answers with noisy statistical decisions. A seeded latent linear SEM based on the same reference graph gives an integration result: at $N = 5,000$, FCI⁺ returns an extra $X \leftrightarrow Y$ with exact-edge F1 0.923 while standard FCI recovers the exact PAG, because the sample CI decisions do not expose the FCI⁺ D-SEP candidate; at $N = 50,000$ both recover the exact PAG. Logical query counts are 102 and 52 at $N = 5,000$, and 102 and 63 at $N = 50,000$, for FCI and FCI⁺ respectively. This is a reproducible fixture, not a universal sample-size threshold.

The application safeguard was also tested on 15 deterministic latent-SEM cases: five realistic graph families, three seeds each, 2,500 observations per case, with a known oracle PAG.

Table 5: Mean recovery across 15 finite-sample oracle cases.

Profile	Skeleton F1	Exact F1	Semantic F1	Endpoint acc.	CI calls
Paper FCI ⁺	0.979	0.536	0.951	0.744	197.1
Robust FCI ⁺	0.985	0.730	0.985	0.840	401.9

The robust profile improves the mean skeleton and endpoint-oriented scores on these runs while using about 2.04 times as many CI calls. This does not prove universal dominance, but it is a known-truth check that the STAR stability gain is not obtained by simply discarding useful structure. It also warns against assuming that fewer CI tests imply uniformly better finite-sample accuracy: FCI and FCI⁺ can make different errors because they take different logical search routes after the initial skeleton.

Key result. The oracle tests support the correctness of the implemented FCI and FCI⁺ search-and-orientation pipeline under exact conditional-independence information. The finite-sample tests simultaneously show that empirical results remain limited by CI-test error and tuning choices.

6 Comparison with Established Software

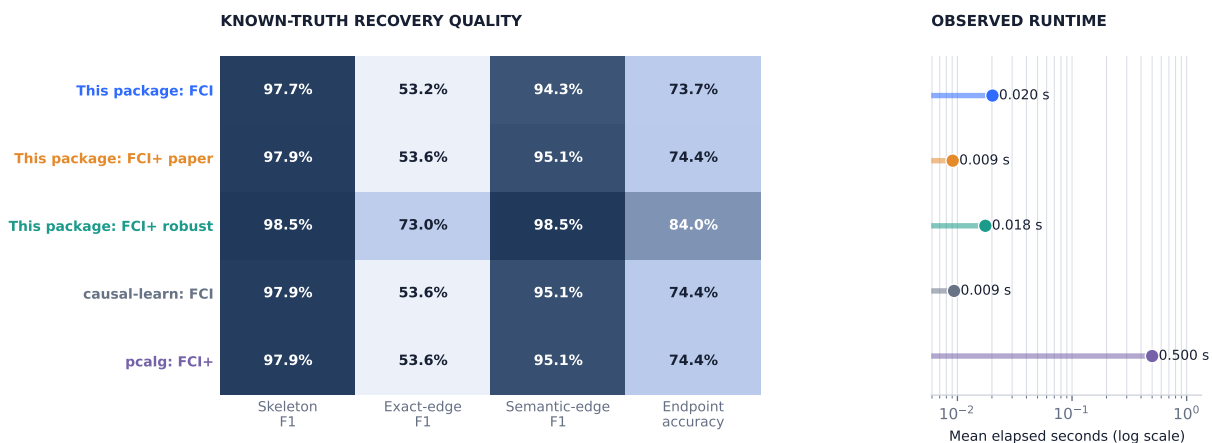
The comparison separates execution from interface inspection. The executed layer asks what happened on identical committed datasets under recorded settings; the capability layer asks what each public interface or official documentation exposes. Tetrad 7.6.10 appears only in the latter as a documentation-only comparison, because no working Java runtime was available in the recorded environment; no empirical row or runtime is attributed to it. All claims are dated 26 July 2026.

The benchmark contains five realistic latent-SEM scenarios, three deterministic repetitions each, and 2,500 observations per run, with every method receiving the same numeric matrix and variable order. Fixed paper and reference rows use Fisher-Z at $\alpha = 0.001$; robust FCI⁺ uses its documented sample-size rule, which resolves to $\alpha = 0.01$ here. The mismatch is intentional because that row evaluates the public robust policy, not a literal Algorithm 2 reproduction.

Table 6: Executed known-truth comparison: means over 15 completed runs.

Method (version)	Skeleton F1	Exact F1	Semantic F1	Endpoint acc.	Seconds	Logical CI
Local FCI paper (0.1.0)	0.977	0.532	0.943	0.737	0.020	449.7
Local FCI ⁺ paper (0.1.0)	0.979	0.536	0.951	0.744	0.009	197.1
Local FCI ⁺ robust (0.1.0)	0.985	0.730	0.985	0.840	0.018	401.9
causal-learn FCI (0.1.4.7)	0.979	0.536	0.951	0.744	0.009	not exposed
pcalg::fciPlus (2.7-12)	0.979	0.536	0.951	0.744	0.500	not exposed

EXECUTED | 15 seeded known-truth cases per method



Descriptive, not a universal ranking: profiles and alpha policies differ by design. The pcalg timing includes R process startup; other timings are in-process calls.

Figure 3: Matched known-truth comparison. The robust profile improves the reported endpoint-oriented means on these scenarios while retaining a similar skeleton score and spending more logical CI queries than paper FCI⁺.

Exact F1 requires both endpoint marks to match; semantic F1 additionally credits endpoint patterns compatible with the target ancestral constraints; endpoint accuracy scores individual marks. The common conversion represents each PAG edge as an unordered node pair plus two semantic marks and never

interprets raw package matrices as generic directed edges. The equality of the aggregate local paper-FCI⁺, causal-learn, and pcalg scores is useful differential evidence on these fixtures, not an equivalence proof: local and causal-learn times cover in-process calls whereas the R timing includes process startup, and `pcalg::fciPlus` does not expose an identical control for the true-degree bound k .

Every comparison obeys five rules. Circles, tails, and arrowheads are preserved in semantic normalisation. Data, order, alpha, CI semantics, and search caps are held fixed where interfaces allow, and every mismatch is disclosed. Standard FCI, Claassen FCI⁺, and other FCI-family hybrids are not relabelled as one algorithm. Logical queries, cache hits, executed statistical tests, and wall time remain different quantities. Smoke-call success establishes callability, not scientific validity. On the capability layer, `pcalg` exposes both FCI and FCI⁺ and accepts user-defined CI tests, and its standard-FCI result exposes separator information, but its inspected `fciPlus` result carries materially less separator provenance; `causal-learn` exposes standard FCI and named CI methods but no inspected public Claassen FCI⁺ entry point; Tetrad provides FCI and several FCI-family or hybrid searches, none of which was established as a drop-in Claassen FCI⁺.

Key result. The local package integrates paper-profile control, structured CI and orientation traces, separator-source provenance, D-SEP diagnostics, per-edge explanations, stability auditing, and bundled artifacts in one Python workflow, which was not found as a single public workflow in the checked interfaces. This is an auditability and integration improvement, not a universal accuracy or speed superiority claim.

7 STAR Study Design and Methods

7.1 Experiment, cohort, and panels

The Tennessee Student/Teacher Achievement Ratio project was a large randomized experiment in kindergarten through grade 3, assigning students and teachers to small classes, regular classes, or regular classes with a full-time aide; official documentation describes small classes of approximately 13–17 students and regular classes of approximately 22–25 (Word et al., 1990). Subsequent economic analysis has used STAR as a central source of evidence on class size (Krueger, 1999). This analysis uses the official Harvard Dataverse release (Achilles et al., 2008): 11,601 student records, 379 variables, CC0 1.0, with codes checked against the accompanying 146-page user guide.

Randomization already provides a stronger basis for estimating the class-size effect than an unconstrained structure-learning algorithm, so FCI is not used to rediscover it. The purpose is to examine whether grade-3 observation is structurally related to early achievement or background variables; whether early and later achievement remain adjacent after conditioning on measured context; how the learned PAG changes when an early post-assignment outcome is included or excluded; whether FCI and FCI⁺ agree on skeletons and important adjacencies; and how much search work FCI⁺ saves on real data. The design also supplies an unusually useful external audit: if a data-only PAG implies an implausible pre-treatment direction, or suggests latent confounding of a genuinely randomized assignment, the discrepancy should be reported rather than accepted literally.

The analysis begins with the 6,325 students who have a kindergarten class assignment, across 79 kindergarten schools, and builds three panels that are deliberately not pooled because each encodes a different statistical question.

Table 7: STAR analysis panels.

Panel	Rows	Nodes	Purpose
Attrition / observation	5,744	9	Relate assignment, characteristics, context, and kindergarten achievement to whether both grade-3 scores are observed.
Longitudinal achievement	2,787	9	Study structure between kindergarten and grade-3 achievement among complete cases while retaining early achievement.
Focused treatment-outcome	2,976	8	Study the class-assignment/grade-3 relationship without allowing kindergarten achievement, an early post-assignment outcome, to serve as a separator.

All model inputs are integer-coded discrete categories: class arm, sex, ethnicity (with smaller source categories collapsed to *Other* to avoid extremely sparse contingency cells), free-lunch status, school context, entry age in quantile categories, binned teacher experience, and reading-plus-mathematics scaled scores converted to achievement quartiles. The exact numeric tables supplied to the algorithms are committed under `case_studies/tennessee_star/data/processed`.

7.2 Analysis protocol

Before discovery, the report computes observed means and randomized-arm contrasts for the kindergarten score sum, the grade-3 score sum among students with both scores, and the probability that both grade-3 scores are observed, with uncertainty from 1,000 bootstrap samples of kindergarten schools. Resampling schools rather than students preserves the main within-school dependence. These contrasts are reported separately from FCI output because they use the experiment’s assignment as the basis for interpretation.

Discovery uses the likelihood-ratio G-square test, $G^2 = 2 \sum_s \sum_{x,y} O_{xys} \log(O_{xys}/E_{xys})$, with zero-count contributions omitted. The test suits the discretised panels, but high-order conditioning can create sparse cells.

Table 8: Primary STAR discovery configuration.

Setting	Value
CI test	G-square on discretised variables
Significance threshold	$\alpha = 0.05$
Standard FCI	profile="paper"; unbounded paper schedule
Paper FCI ⁺	profile="paper"; $k = 3$
Robust application FCI ⁺	profile="practical"; stable skeleton; strongest separating set; conservative colliders; robust orientation; depth and sparsity bound 3
R reference	pcalg::fciPlus; selection bias enabled; no exposed explicit k
Timing	Three full-data repetitions; median elapsed time reported
PAG stability	100 kindergarten-school cluster bootstrap fits per algorithm and panel
Order audit	Every cyclic column shift for all local profiles and the R reference
Prior temporal constraints	None supplied to discovery; temporal reversals are audited after fitting

For stability, the 79 schools are sampled with replacement, all student rows from a selected school are included, the algorithm is refitted, and each unordered adjacency is recorded; with $B = 100$ replicates the estimated frequency is $\hat{\pi}_{XY} = B^{-1} \sum_b \mathbf{1}\{X, Y \text{ adjacent in replicate } b\}$. This is an exploratory stability diagnostic with one-percentage-point resolution, not a confidence interval, a posterior edge probability,

or an endpoint-compatible uncertainty summary. The focused analysis is additionally repeated for $\alpha \in \{0.01, 0.05\}$, three versus four quantile bins, and FCI+ bounds $k \in \{2, 3, 4\}$.

The application distinguishes a reproducible adjacency from a credible direction. A skeleton-level conclusion is promoted only when the target adjacency survives every cyclic column shift and has high bootstrap frequency. A directional conclusion additionally requires invariant endpoint marks, no conflict with measurement time or randomization, and no material reversal under the sensitivity checks. Failure of the directional rule does not delete the edge; it downgrades the claim to conditional-dependence structure with unresolved direction.

8 STAR Results

8.1 Observed randomized-arm outcomes

Table 9: Observed outcome summaries by kindergarten class arm. Intervals use 1,000 kindergarten-school bootstrap samples.

Outcome	Arm	<i>n</i>	Estimate	95% interval
Kindergarten score	Regular	2,005	918.04	908.78 to 927.05
	Regular + aide	2,043	918.36	908.52 to 927.07
	Small	1,738	931.94	922.70 to 940.12
Grade-3 score	Regular	1,047	1246.74	1238.96 to 1254.01
	Regular + aide	1,019	1247.25	1238.53 to 1255.48
	Small	936	1258.43	1250.58 to 1266.17
Grade-3 observed	Regular	2,194	47.72%	42.72% to 52.14%
	Regular + aide	2,231	45.67%	40.64% to 50.39%
	Small	1,900	49.26%	44.46% to 53.77%

Grade-3 summaries condition on both scores being observed. Kindergarten summaries also exclude missing score components, so both are design-supported observed-outcome comparisons rather than a fully reconstructed intention-to-treat estimand.

Randomized-arm descriptive contrasts with 95% school-cluster intervals

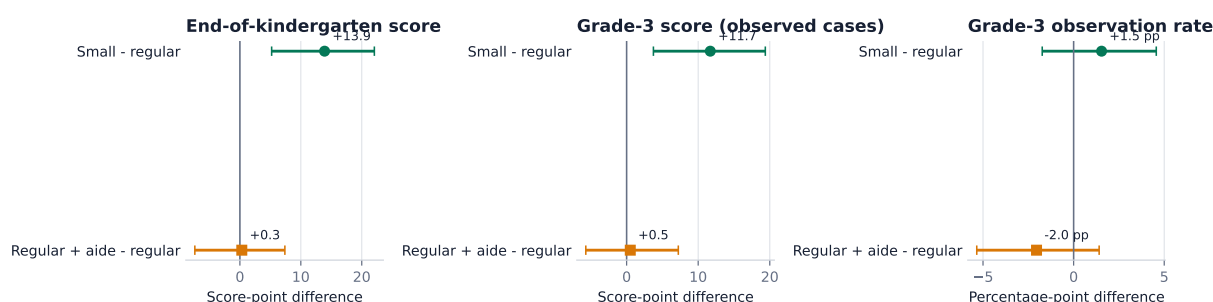


Figure 4: Contrasts against the regular-class arm. Small classes show positive score differences at kindergarten and grade 3 among observed cases; the aide arm remains close to zero.

The small-minus-regular kindergarten contrast is 13.90 points, with a 95% school-cluster interval from 5.21 to 22.07, while the regular-with-aide contrast is 0.31 points (−7.40 to 7.38). At grade 3 among observed students the small-class contrast remains positive at 11.69 points (3.74 to 19.38) and the aide contrast is 0.51 points (−5.70 to 7.22). Observation rates show no precise arm difference: small minus regular is +1.54 percentage points (−1.73 to 4.56) and aide minus regular is −2.05 points (−5.35 to 1.41).

8.2 Attrition and longitudinal panels

Attrition / observation

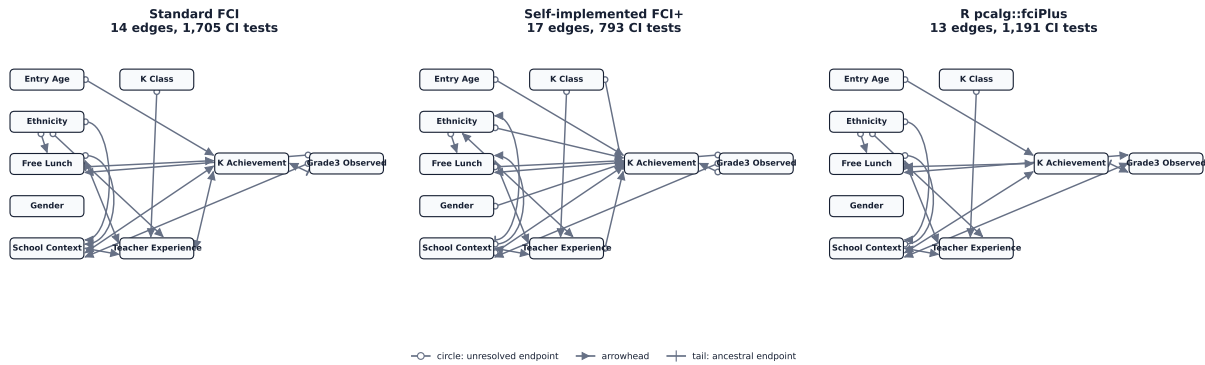


Figure 5: Three PAGs for the attrition/observation panel: self-implemented FCI, self-implemented FCI⁺, and R `pcalg::fciPlus`. The skeletons substantially overlap, while the achievement/observation endpoint differs.

On the attrition panel, standard FCI returns 14 edges after 1,705 CI tests, self-implemented FCI⁺ 17 edges after 793 tests, and R `pcalg::fciPlus` 13 edges after 1,191 tests; pairwise skeleton Jaccard values are 0.824, 0.929, and 0.765. All three omit a direct `K_Class`–`Grade3_Observed` adjacency, which appears in only 12–16% of the school bootstraps across local profiles. By contrast, `K_Achievement`–`Grade3_Observed` is adjacent in 100% of bootstrap fits for all three local profiles, and the free-lunch/observation adjacency is also stable. Later data availability is therefore not independent of early achievement and measured socioeconomic or school context. The three endpoint readings are revealing: standard FCI gives the impossible *grade-3 observation* → *kindergarten achievement*; self-FCI⁺ leaves a circle at the later endpoint; and R-FCI⁺ gives the temporally admissible direction. The R result is more plausible on this edge, but package identity alone cannot establish that its orientation is statistically correct.

Longitudinal achievement

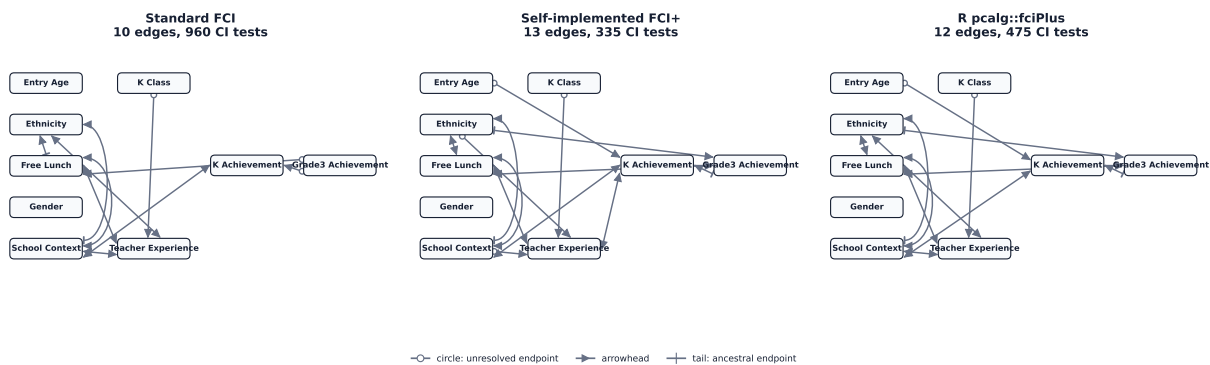


Figure 6: Three PAGs for the longitudinal complete-case panel. All retain the kindergarten/grade-3 achievement adjacency, but both FCI⁺ implementations orient it against chronology.

On the longitudinal panel, standard FCI returns 10 edges and 960 tests, self-FCI⁺ 13 edges and 335 tests, and R-FCI⁺ 12 edges and 475 tests, with skeleton Jaccard 0.769, 0.833, and 0.923. All three retain the kindergarten/grade-3 achievement adjacency, which appears in all 100 bootstrap samples for all three local profiles. Neither full-data PAG retains a direct class-assignment/grade-3 adjacency once kindergarten achievement is included, yet that adjacency reappears in 71% of FCI, 79% of paper FCI⁺, and 80% of robust-profile bootstrap fits — a warning that conservative orientation and order invariance do not remove school-resampling sensitivity. Both FCI⁺ implementations orient grade-3 achievement toward kindergarten

achievement, which is temporally impossible if read causally, while standard FCI leaves that endpoint unresolved. Agreement between two implementations is therefore not sufficient to override chronology: they may replicate the same finite-sample CI pattern and orientation logic under complete-case selection.

Interpretation caution. The attrition PAG does not prove a particular missingness mechanism. It shows that the observed conditional-independence structure links grade-3 availability to earlier achievement and context. A formal missing-data analysis would require an explicit estimand and additional assumptions.

8.3 Focused treatment-outcome panel

Focused treatment-outcome

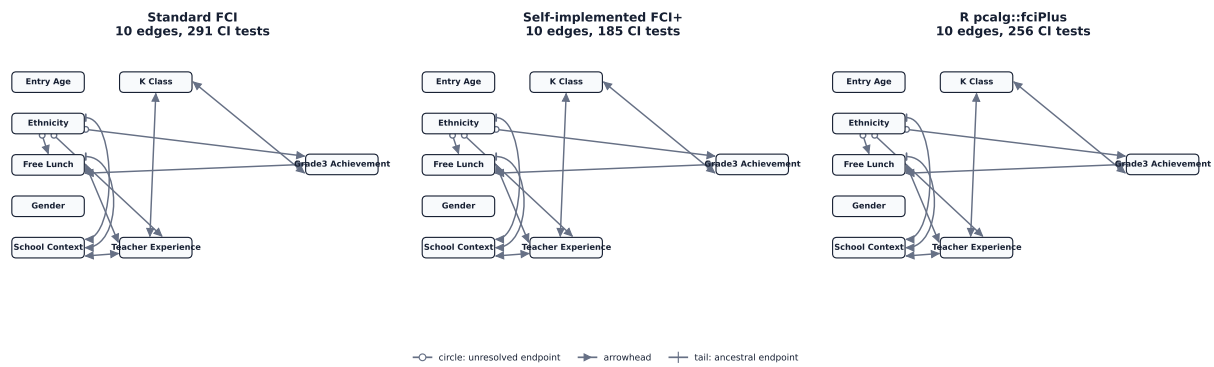


Figure 7: PAGs for the focused complete-case panel. All three algorithms recover the exact same 10-edge skeleton and endpoint pattern, including the bidirected class-assignment/grade-3 edge.

Removing kindergarten achievement from the node set changes the result. Standard FCI, self-FCI⁺, and R-FCI⁺ all return the same 10-edge PAG — not merely the same skeleton — and all three contain $K_Class \leftrightarrow Grade3_Achievement$. The adjacency appears in 92% of FCI, 85% of paper self-FCI⁺, and 96% of robust-profile bootstraps. Independent exact replication makes this observed conditional-independence pattern more credible, but it does not make a literal latent-confounding reading credible: kindergarten class assignment was randomized, while the focused panel conditions on grade-3 outcomes being observed and omits early post-assignment achievement. Conditioning on observation, school-level dependence, omitted post-randomization variables, and CI-test approximation can all create an observed pattern represented by arrowheads at both endpoints.

Sensitivity of the focused class-assignment / grade-3 adjacency

	Absent 225 CI tests	Present 305 CI tests	Absent 252 CI tests	Present 291 CI tests
Standard FCI				
FCI+ paper, $k=2$	Absent 146 CI tests	Present 203 CI tests	Absent 152 CI tests	Present 174 CI tests
FCI+ paper, $k=3$	Absent 150 CI tests	Present 214 CI tests	Absent 163 CI tests	Present 185 CI tests
FCI+ paper, $k=4$	Absent 150 CI tests	Present 214 CI tests	Absent 163 CI tests	Present 185 CI tests
FCI+ robust, $k=2$	Absent 217 CI tests	Present 316 CI tests	Absent 223 CI tests	Present 290 CI tests
FCI+ robust, $k=3$	Absent 233 CI tests	Present 360 CI tests	Absent 253 CI tests	Present 334 CI tests
FCI+ robust, $k=4$	Absent 234 CI tests	Present 367 CI tests	Absent 257 CI tests	Present 341 CI tests
	3 bins $\alpha = 0.01$	3 bins $\alpha = 0.05$	4 bins $\alpha = 0.01$	4 bins $\alpha = 0.05$

Figure 8: Focused treatment-outcome sensitivity. The adjacency appears at $\alpha = 0.05$ and disappears at $\alpha = 0.01$, regardless of whether three or four quantile bins or $k = 2, 3, 4$ is used for FCI+.

The sensitivity result is decisive for interpretation. The focused adjacency is robust to the tested bin count and degree bound but not to the CI threshold: at $\alpha = 0.01$ no local configuration retains it, while at $\alpha = 0.05$ the paper profiles retain a bidirected mark for every tested k , and the robust endpoint is $K_Class \leftarrow \circ Grade3_Achievement$ with three bins and $K_Class \circ - \circ Grade3_Achievement$ with four. The stricter threshold requires more evidence against conditional independence before an edge is kept, so its removal indicates that the edge rests on relatively weak conditional-dependence evidence rather than an invariant feature across reasonable settings.

8.4 Robust application profile, agreement, and cost

Robust FCI+ application profile: conservative STAR PAGs

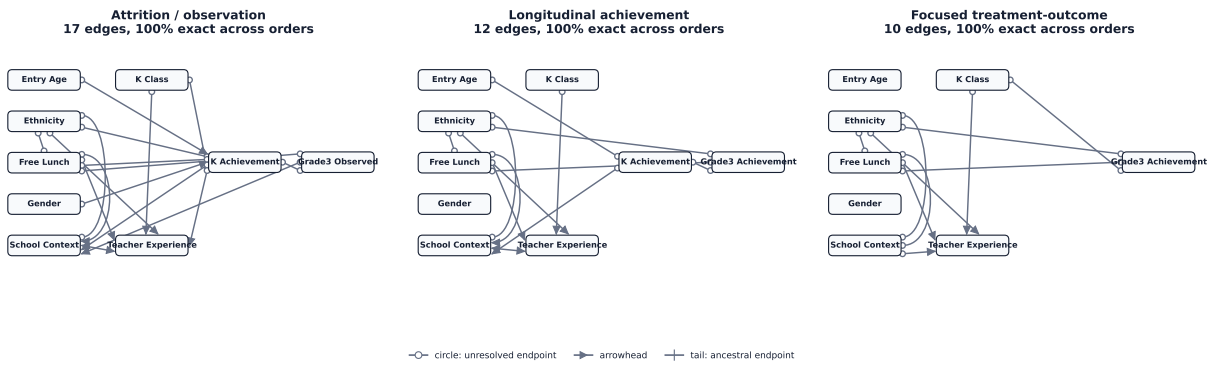


Figure 9: The three STAR PAGs from the separate robust application profile. Stable skeleton search and conservative orientation preserve the main adjacencies while leaving unsupported directions unresolved.

The robust profile is not a fourth paper implementation; it is a finite-sample decision layer applied after the literal comparisons. It returns 17, 12, and 10 edges on the attrition, longitudinal, and focused panels, with target relations $K_Achievement \circ - \circ Grade3_Observed$, $K_Achievement \circ - \circ Grade3_Achievement$, and $K_Class \circ - \circ Grade3_Achievement$. It thus retains the relevant skeleton information without repeating the paper-profile chronology reversal or the focused-panel bidirected over-interpretation.

Table 10: Cyclic-order audit for paper and robust self-FCI⁺ profiles.

Panel	Profile	Exact PAG	Worst skeleton	Temporal reversals
Attrition	Paper	100.0%	100.0%	0
	Robust	100.0%	100.0%	0
Longitudinal	Paper	55.6%	92.3%	1
	Robust	100.0%	100.0%	0
Focused	Paper	75.0%	100.0%	0
	Robust	100.0%	100.0%	0

The order audit rotates the input variable list through every cyclic shift. The R reference preserves every panel skeleton while reproducing exact PAG endpoints in 7 of 9 attrition orders, 6 of 9 longitudinal orders, and 6 of 8 focused orders. Paper self-FCI⁺ is exact in 9 of 9, 5 of 9, and 6 of 8 respectively, and its longitudinal skeleton changes slightly in some orders. Robust self-FCI⁺ is exact in all 26 refits.

Table 11: Three-algorithm structural agreement (left) and search cost (right). Endpoint rates use only adjacencies shared by the indicated pair.

Panel	Pair	Jaccard	Endpoints	Panel	Algorithm	Edges	CI	Sec.
Attr.	FCI / self	0.824	50.0%	Attr.	FCI	14	1,705	5.437
	FCI / R	0.929	61.5%		self paper	17	793	1.788
	self / R	0.765	38.5%		self robust	17	1,304	2.838
					R-FCI ⁺	13	1,191	10.394
Long.	FCI / self	0.769	50.0%	Long.	FCI	10	960	2.148
	FCI / R	0.833	70.0%		self paper	13	335	0.337
	self / R	0.923	83.3%		self robust	12	619	0.562
					R-FCI ⁺	12	475	1.650
Focus.	FCI / self	1.000	100.0%	Focus.	FCI	10	291	0.322
	FCI / R	1.000	100.0%		self paper	10	185	0.159
	self / R	1.000	100.0%		self robust	10	334	0.247
					R-FCI ⁺	10	256	0.884

Paper FCI⁺ uses 46.5%, 34.9%, and 63.6% of the standard-FCI CI-test counts on the three panels, and the independent R reference uses 69.9%, 49.5%, and 88.0%. Runtime is not a language-neutral benchmark, because the R runner includes R-level contingency-table construction; CI-call counts are the more interpretable comparison. Relative to paper self-FCI⁺, the robust profile uses 1.64, 1.85, and 1.81 times as many CI calls, purchasing stable deletion and invariant conservative endpoints rather than speed.

Key result. The robust profile improves the measured reliability of the STAR output: it returns the same full PAG object for every tested cyclic order, every target adjacency survives, and no fully directed edge runs backward in measurement time. The price is deliberate under-orientation and additional CI work. Reliability has improved; identifiability has not.

9 Interpretation, Evidence Hierarchy, and Limitations

9.1 Evidence hierarchy

The application uses an explicit hierarchy so that a visually definite PAG mark cannot outrank the experimental design. Tier 1 is randomized assignment with the school-cluster arm contrast, permitting a

bounded causal statement for the analysed observed outcome, qualified by missingness, compliance, and unreconstructed assignment blocks. Tier 2 is an adjacency replicated across methods, resamples, tested orders, and sensitivity settings, permitting a robust skeleton-level structural claim with no magnitude or direction. Tier 3 is a stable skeleton with circles or endpoint disagreement, where adjacency is more credible than orientation. Tier 4 is an implementation-, order-, or chronology-sensitive endpoint, which is diagnostic evidence only. Tier 5 is a threshold-sensitive or complete-case exploratory pattern, a hypothesis for external validation.

9.2 What can and cannot be concluded

Within the kindergarten cohort and among students with observed score components, the small-class arm is 13.90 points above the regular arm (school-cluster 95% interval 5.21 to 22.07). This design-supported observed-arm contrast is consistent with a beneficial early effect, but it is not an identified causal effect in the post-assignment observed-outcome subset without additional assumptions about missing outcomes, compliance, and the original randomization blocks. The regular-with-aide contrasts are close to zero at both time points with intervals covering zero, so these data show no evidence that adding an aide reproduces the advantage of reducing class size. Kindergarten and grade-3 achievement remain adjacent in the full longitudinal fits of all three algorithms and in 100% of the school bootstraps, and kindergarten achievement is also adjacent to grade-3 observation in all three attrition fits, supporting a robust *skeleton-level* relationship between early performance, later performance, and later availability. The full-data attrition PAGs do not retain a direct class-arm/observation adjacency and its bootstrap frequency is low, so the evidence supports concern about selective attrition without showing a stable direct effect of arm on observation.

Conversely: no unique DAG is identified, since every PAG represents an equivalence class. No causal effect magnitude comes from FCI; PAG edges contain no “points gained” coefficient. A bidirected edge is not proof of a hidden confounder, and in the focused panel it conflicts with the known randomization unless selection or post-treatment conditioning is considered. The grade-3 contrast is not a complete long-run intention-to-treat estimate. Temporal reversals are algorithm diagnostics under violated or approximate assumptions, not credible chronology. Implementation agreement is not proof, because shared CI decisions and shared logical rules can reproduce the same wrong endpoint. The 26 recorded cyclic shifts are a finite audit, not a proof of general order invariance. Bootstrap frequency is not a posterior probability. Finally, absence is policy conditional: failure to retain an edge under one node set, CI test, and threshold is not proof of zero causal effect.

9.3 Limitations

Achievement and age are reduced to quantile categories and teacher experience is binned, which enables G-square testing across mixed educational variables but discards within-bin information and can leave small expected counts in multiway cells as conditioning sets grow. The CI test treats student rows as independent; the bootstrap resamples schools, which measures sensitivity to school composition but does not make each underlying G-square test cluster-robust. The longitudinal and focused panels require grade-3 scores, so conditioning on complete cases may open selection paths — a concern reinforced by the attrition panel’s own findings. Kindergarten achievement and teacher experience are measured during the treatment period, so including the former may block part of the class-size pathway while excluding it may leave omitted post-treatment dependence; both specifications are reported rather than selecting the one with the preferred edge. FCI-family guarantees rely on faithfulness and correct CI decisions, and the threshold-sensitive focused edge is a direct example of that limitation. The true maximum degree of the STAR MAG is unobserved, so $k = 3$ is a bounded analysis setting rather than a verified assumption, and `pcalg::fciPlus` does not expose a comparable argument. The package is version 0.1.0 and classified pre-alpha; hand-authored oracle targets are strong regression evidence, not an independent formal proof or an exhaustive MAG-equivalence checker, and the complete Figure 4(b) PAG target is repository derived even though its MAG and exhibited separator are source grounded.

A deterministic scaling audit runs the two local paper profiles on a directed chain and on the Figure 4(b) D-SEP core augmented with isolated variables, at 5, 10, 20, 40, and 80 nodes; every run recovers the exact target skeleton. On the five-node core, self-FCI⁺ uses 62 queries versus 101 for standard FCI, but that advantage does not persist once irrelevant isolated variables are added: at 80 nodes the counts are 5,162 versus 3,251, and on the 80-node chain, which has no difficult D-sep link, 24,338 versus 12,638. This does not contradict the fixed- k theorem, which is a worst-case upper bound rather than a pairwise dominance result; augmentation has a real cost when targeted D-SEP search offers little benefit. Substantively, original randomization blocks, class switching, compliance, and school-specific assignment procedures are not reconstructed, and the transparent composite score simply adds reading and mathematics, leaving test scaling, subject-specific, distributional, and subgroup effects unexamined.

10 Conclusions and Future Work

Source fidelity is explicit and testable. Standard FCI maps to the historical 2000 schedule, whose Theorem 6.4 is a correctness result for a partially oriented inducing-path graph under correct CI information, not a modern maximal-orientation theorem. FCI⁺ maps to the augmented skeleton and hierarchy of Algorithm 2, whose polynomial result remains the restricted $O(N^{2(k+2)})$ exact-oracle query bound for fixed true degree k , faithfulness, and constant-time CI. The concept/symbol/test mapping makes literal paper mechanisms, later orientation completeness, neutral software structure, and robust engineering policy visibly different.

Validation is layered rather than circular. Exact m-separation tests recover the independently specified targets on the encoded fixtures, including hierarchical removal of the Figure 4(b) false link with the source-exhibited separator $\{U, V, Z\}$. The known-truth benchmark executes five methods over 15 runs: robust FCI⁺ improves the recorded mean endpoint metrics, while paper FCI⁺ reduces logical queries relative to standard FCI. The causal-learn and `pcaIlg` comparisons supply differential evidence and Tetrad remains documentation-only. The package's defensible improvement is its integrated audit surface, not a claim to be universally fastest or most accurate.

The package was then applied to a real educational experiment without mixing domain code into the algorithm. STAR shows a clear observed small-class advantage, no comparable aide advantage, stable early-to-later achievement structure, and evidence that later observation is selective. An independent R run exactly replicates the focused 10-edge PAG and broadly supports the other skeletons, while also showing why PAGs require disciplined interpretation: the most tempting bidirected treatment-outcome edge is conditional on complete cases and disappears under a stricter alpha, and some endpoint marks change with variable order or violate chronology. The separate robust profile addresses the principal empirical failure found by those audits, returning the same PAG across every tested cyclic order in all three panels, preserving the main target adjacencies, and reporting no fully directed temporal reversal — at the cost of unresolved endpoints and more CI calls.

No algorithm is uniformly best on these data. Paper self-FCI⁺ is the fidelity and query-efficiency subject; robust self-FCI⁺ is preferred for applied interpretation under the predeclared audits although its circles identify less direction; R `pcaIlg` provides valuable independent replication and the most plausible attrition direction, yet shares the longitudinal reversal and shows endpoint order sensitivity; standard FCI is less economical but leaves the longitudinal target partly unresolved rather than fully reversed. Credibility must be assigned claim by claim. The report therefore rejects two extremes: it is too pessimistic to conclude that causal discovery adds nothing to a randomized study, since the attrition and stability analysis reveals useful structure, and too optimistic to conclude that FCI has rediscovered and fully identified the experiment's causal model. The correct contribution lies between: an auditable map of conditional-independence constraints, uncertainty, and modelling sensitivity.

Intended use. The package is built for three audiences. Applied researchers obtain latent-confounding-aware structure discovery in which every edge can be traced to the CI decisions, separator provenance, and orientation events that produced it — the property that makes the STAR audits possible and that section 6

did not find as one public workflow in the checked interfaces. Methodologists obtain a typed, paper-profile-controlled reference implementation against which other FCI-family software can be differentially tested, in an ecosystem where the inspected Python interfaces expose no public Claassen FCI⁺ entry point. Instructors and reviewers obtain a worked, fully reproducible template for keeping design-based evidence and discovery evidence separate in a randomized study.

Recommended extensions are to reconstruct the randomization strata and estimate formal intention-to-treat effects with an explicit model for grade-3 observation; develop cluster-aware or multilevel CI tests; report Monte Carlo uncertainty for the stability frequencies and extend stability summaries to endpoint compatibility; analyse reading and mathematics separately with prespecified heterogeneity hypotheses; evaluate defensible temporal background knowledge separately from the data-only robust profile; compare alternative mixed-data CI tests; and add an independent exhaustive MAG-equivalence oracle for target-PAG derivation.

Reproduction. The complete STAR application is rebuilt by running `case_studies.tennessee_star.run_case_study` with `--benchmark-repeats 3`, `--bootstraps 100`, and `--descriptive-bootstraps 1000`; adding `--refresh-pcalg` re-executes the independent R reference. The scripts `generate_report_figures.py`, `generate_oracle_validation_summary.py`, and `generate_scaling_benchmark.py` under `reports/` regenerate the figures, the frozen Figure 4(b) evidence, and the scaling audit. Quality gates are Pytest, Ruff, strict MyPy over both the package and the case study, and a wheel build with an installed-import smoke test. The committed `STAR_Students.tab.gz` has SHA-256 `2ffa578822eb30fcd6626fa7c5cc734a721fd05dd522403f0d643f89e891bac8`.

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